Project Proposal

A Hybrid Transformer–Mamba Architecture for Academic Text Style Enhancement

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Project goal. Develop an academic-writing assistant that detects weakly academic expressions and rewrites them into clearer, more professional language while preserving meaning. The **Transformer–Mamba hybrid** is our proposed model; Transformer-only and Mamba-only systems are baselines.

1. Project Idea

Academic writing often contains conversational wording, vague claims, unnecessary intensifiers, or structurally weak sentences. We propose a two-stage system: **(1) style detection** to identify sentences or spans needing revision and **(2) neural rewriting** to produce a more formal academic version without changing the intended meaning. The core rewriter combines a Transformer encoder for global semantic relationships with Mamba blocks for efficient selective state-space sequence modeling. Their representations are merged through residual or gated fusion before a Transformer-style decoder generates the final rewrite.

Input → Detect weak style → Transformer encoding → Mamba refinement → Feature fusion → Academic rewrite

2. Implementation Plan

Data and preprocessing

- Build paired *ordinary/informal* → *academic* examples from suitable public academic revision/style-transfer resources.
- Supplement coverage with carefully generated synthetic pairs, followed by filtering and manual spot-checking.
- Clean, deduplicate, normalize, tokenize, and create train/validation/test splits; prepare a smaller academic vs. non-academic set for detection.

Evaluation metrics

- **BLEU / ROUGE-L**: reference overlap.
- **BERTScore**: semantic preservation.
- **Academic-style accuracy**: improvement in academic tone.
- Qualitative review of fluency, meaning preservation, and over-rewriting; small human evaluation if time permits.

Proposed model and training

- Transformer encoder + Mamba blocks + residual/gated fusion + Transformer decoder.
- Fine-tune pretrained components where feasible and optimize with sequence-to-sequence cross-entropy.
- Use validation loss/early stopping and matched settings for fair comparison with Transformer-only and Mamba-only baselines.

Experiments and analysis

- Main study: Transformer-only vs. Mamba-only vs. proposed Hybrid.
- Ablation: no fusion / residual fusion / gated fusion; optionally vary Mamba depth.
- Report parameter count, inference time, peak memory, representative outputs, and failure cases.

3. Group Member Responsibilities

Member	Primary responsibilities
Borui Ning	Lead model architecture; implement the Transformer–Mamba hybrid and feature-fusion mechanism; coordinate model integration and major design decisions.
Licheng Wang	Collect/construct paired rewriting data; clean, deduplicate, preprocess, split the dataset, and prepare labels for the style detector.
Haoxiang Ning	Build training/evaluation pipelines; implement baselines; perform hyperparameter tuning, metrics, ablation studies, and quantitative error analysis.
Lingfeng Lu	Integrate the inference prototype/interface; conduct qualitative testing and visualization; organize experiment outputs and support final documentation/report preparation.

4. Expected Output and Success Criteria

Prototype	A working assistant that flags weakly academic expressions and returns a more formal rewrite while preserving the source meaning.
Model study	A reproducible comparison of Transformer-only, Mamba-only, and the proposed Transformer–Mamba hybrid, including fusion ablations and efficiency measurements.
Success criteria	Better academic-style scores without unacceptable semantic degradation, supported by automatic metrics, representative outputs, and failure-case analysis.

Submission: One member will submit this PDF on behalf of the entire group. **Format:** PDF, maximum one page.